

Residency Block Schedule — Master Rule Set

The authoritative, plain-English list of the rules a valid block schedule must follow. Maintained for collaborators and for hand-auditing a schedule (including rules the solver does **not** flag). Last updated 2026-06-18.

How to use this document. Each rule states what must be true, and an **Enforcement** tag saying where it is guaranteed: -  **Solver-hard** — the optimizer cannot produce a schedule that violates it. -  **Soft objective** — the optimizer is *pushed* toward it but may trade it off; worth checking by hand. -  **Operational / manual** — **not modeled in the solver at all**; it is satisfied by external staffing (BMC, TY residents) or must be checked by a human. **These are the rules most likely to be silently violated — audit them first.**

0. Terminology (read this first)

- The program's **block = 4 weeks**. A **half-block = 2 weeks**.
- The academic year = **52 weeks = 13 blocks = 26 half-blocks**.
- Internally the app/code calls each 2-week half-block a "block" (a 26-unit grid, "slots" in this document). So **1 program block = 2 slots**. Slot index = position 0–25; slot label = e.g. **1A** (slot 0), **1B** (slot 1), ... **13B** (slot 25).
- **Block 7** = slots 12–13 (labels 7A, 7B). **Block 13** = slots 24–25 (13A, 13B).
- The schedule covers the **11 categorical PGY-1 interns**. Other residents (**TY** = transitional-year; **BMC** = Broadlawns group) also appear on some rotations and count toward coverage, but the solver does **not** schedule them — their placement is external/variable.

1. Per-block rotation capacities (hard limits)

How many people may be on a rotation in any given block. "Categorical" = the 11 interns; "total" = categoricals + TY + BMC together. **These are real bed/slot limits — exceeding them is a true violation even if a schedule otherwise looks fine.**

Rotation	Rule	Enforcement
ICU	≤ 1 categorical per block; ≤ 2 total (a TY may add a 2nd body)	 Solver-hard
VA	≤ 2 categoricals per block (never 3)	 Solver-hard
Broadlawns (BMC rotation)	≤ 2 total (categorical + TY) per block	 Solver-hard (total cap)
Yunker 7 Days	exactly 2 total every block, never more; categoricals ≤ 1 per block except block 13 = exactly 2; never 2 BMC together (see §2)	 Solver-hard + 
Yunker 8 Pulmonology	≥ 1 every block (continuous coverage); evenly distributed	 / 
All other rotations	per-rotation <code>max_per_week</code> from config	 Solver-hard

Audit tip: count bodies per column (block) for ICU, VA, BMC, Y7 Days. ICU/VA/BMC are quick: never more than the number above. Y7 Days is special — see §2.

2. Younger 7 Days coverage (the most nuanced rule)

Y7 Days must have exactly 2 people every block — never 1, never more than 2. The two bodies are composed from three sources, and the composition varies by block:

Block(s)	Body 1	Body 2
Most blocks (a categorical is placed)	1 categorical	1 BMC
Blocks where no categorical is placed (e.g. 3, 6, 12)	1 BMC	1 TY
Block 7 (slots 12–13)	1 categorical	1 TY (BMC is absent this block)
Block 13 (slots 24–25)	1 categorical	1 categorical (no BMC, no TY)

Rules that produce this: - Each categorical does **exactly one** 4-week Y7 Days block during the year — this is an **absolute requirement; a categorical must never lose their Y7 Days block.** ■ - BMC supplies **exactly ONE body** on every block **except block 7 and block 13** (where it is absent). **Never two BMC on Y7 Days at once.** ■ (written into the schedule by the app's filler so it is **visible**, not assumed) - TY supplies the **2nd body** on any block where a categorical is not placed (and at block 7). TY placement is variable/external; the app writes a TY placeholder so the schedule visibly shows the full team of 2. ■ - **Block 13 = 2 categoricals** (solver-hard). ■ - **Total is capped at 2** — never 3. ■

Audit tip: every Y7 Days block should show exactly 2 names. Valid combinations: categorical+BMC, BMC+TY, categorical+TY (block 7 only), or categorical+categorical (block 13 only). **Invalid:** any block with 1 or 3 bodies, or 2 BMC together, or a categorical without their one required Y7 Days block somewhere in the year.

3. Younger 7 Nights (call-driving rotation)

- **Always two separate 2-week segments** — Y7 Nights is **never** a single 4-week run. The two segments are split across the year. ■
- A resident's two night segments may be close (2 weeks apart) or far (up to opposite ends of the year); both are acceptable. ■ (well-distributed, not crammed)
- Y7 Nights cannot start in the first program block (earliest start enforced). ■
- A break is required between the two segments. ■

4. The Elective / Younger 7 Nights link rule

- Each categorical does **2 slots total** of (Younger 7 Nights + Elective) combined. ■
- Only **2 Elective slots exist program-wide** per year. ■
- Concretely: **9 of 11** residents do 4 weeks of Y7 Nights; **2 of 11** do 2 weeks of Y7 Nights + 2 weeks of Elective. ■

Audit tip: exactly 2 residents should have an Elective block; everyone else has the full 4 weeks of nights.

5. Younger 7 weekend night call coverage (the program's core purpose)

The block schedule exists so the downstream **call schedule** can cover Younger 7 weekend nights without resorting to volunteers. A weekend belongs to the rotation a resident is on (a block runs Mon → 2nd Sunday).

- **Saturday Y7 night** is covered by the intern on **Younker 8 Pulmonology**. With 11 interns × 2 half-blocks each = 22 of 26 half-blocks, exactly **4 half-blocks have no intern on Y8 Pulm** — these are covered by upper-levels and are **mathematically unavoidable**, not a defect. 🟡 operational
- **Sunday Y7 night** priority chain: (1) Inpatient GI intern → (2) Infectious Disease intern → (3) any intern on an outpatient/elective rotation → (4) volunteer (last resort). 🟡 operational
- **Eligibility rule (critical for auditing)**: an intern can cover a given weekend only if they are **not on a heavy rotation that block and not entering a heavy rotation the next block** (a rested-weekend rule — they can't take call the weekend before a heavy Monday start). 🟢 soft objective (the solver is pushed to keep ≥2 eligible coverers per weekend; **0 means a volunteer is forced**).

Audit tip: for each weekend, list interns who are (on GI/ID/light now) AND (not entering a heavy rotation next block). Zero such interns = a volunteer weekend = the thing the schedule is meant to prevent.

6. Workload tiers & transition quality

Used for judging how punishing a resident's run of rotations is. **The app's internal `rotation_type` flag is unreliable for this — use this list.**

- **Heavy (6):** ICU, VA, Broadlawns, Younker 7 Days, Younker 7 Nights, Younker 8 Pulm.
- **Medium / consult (2):** Inpatient GI, Infectious Disease (lighter; PTO-eligible — this is why they carry call).
- **Light (8):** Outpatient GI, Outpatient Pulm, Ambulatory A, Emergency Medicine, Addiction Medicine, Elective, Outpatient TIC Cardiology, Outpatient UPH Cardiology.

Transition rules (measured in consecutive weeks of heavy + medium): 🟢 soft - **6 weeks** = preferred maximum. - **8 weeks** = acceptable but not ideal. - **10+ weeks** = avoid (e.g. 4wk VA → 2wk ID → 4wk ICU). - Avoid jumping directly from one heavy rotation into a **different** heavy rotation. - The bad pattern to avoid is *stacked heavies* (VA→ID→ICU). A run that alternates heavy with medium/light (e.g. Y7 Nights → GI → Y7 Nights → ID) is far gentler even if long.

7. Rotation shapes & structure

- **4-week full blocks, even-start, consecutive:** ICU, Broadlawns, Younker 7 Days. 🟡
 - **Flexible 2-or-4 weeks:** VA, Younker 8 Pulmonology, Ambulatory A. 🟡
 - **2-week half-blocks only:** everything else. 🟡
 - VA requires a break between its segments. 🟡
 - Each resident is scheduled **every block** (52/52 weeks); there is no vacation/flex in the model — intentional. 🟡
 - Rotations are not always exactly 4 weeks because 13 blocks span 52 weeks; blocks 1 and 13 carry edge-case nuances (e.g. Y7 Days block 13 staffing in §2).
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8. Per-resident requirements

- Each categorical completes their required rotations for the year (minimums per rotation). 🟡 (Note: *historically these minimums were under-enforced for VA / Outpatient GI / Ambulatory A / Outpatient Pulm — see RULES_REVIEW.md item B1. Verify a re-solve respects them.*)
- No prerequisite-rotation ordering is required. 🟡 (none configured)

- Inpatient blocks are spaced out where possible (only specific GI/ID adjacency rules are explicitly encoded).



9. What the solver does NOT model (audit these by hand)

- **The call schedule itself** — no call eligibility/credit/coverage is computed here; the block rules are a *proxy* for call-feasibility (§5).
 - **TY resident placement** — variable year-to-year, external.
 - **BMC staffing** beyond the Y7 Days filler — external.
 - **Time off / PTO** — handled in the downstream call system, not here.
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Appendix — quick hand-audit checklist

1. **ICU** ≤ 1 categorical / ≤ 2 total per block.
2. **VA** ≤ 2 categoricals per block.
3. **Broadlawns** ≤ 2 total per block.
4. **Younker 7 Days** = exactly 2 total every block (valid pairs: cat+BMC, BMC+TY, cat+TY at blk 7, cat+cat at blk 13); never 2 BMC together; never 1 or 3; each categorical does exactly 1 Y7 Days block somewhere in the year.
5. **Younker 7 Nights** = two separate 2-week segments per resident; never a 4-week run.
6. **Elective** appears exactly twice program-wide; those 2 residents do 2wk nights.
7. **Every Sunday weekend** has ≥ 1 eligible intern (GI/ID/light now, not entering heavy next) — ideally ≥ 2 .
Zero = a forced volunteer.
8. **No consecutive heavy+medium run > 8 weeks** (preferably ≤ 6); no direct heavy→different-heavy jumps.
9. **Each resident scheduled every block**, completes required rotations.